Python Tuples

Creating a Tuple

```
# empty tuple
my_tuple = ()

# tuple having integers
my_tuple = (1, 2, 3)

# tuple with mixed datatypes
my_tuple = (1, "Hello", 3.4)

# nested tuple
my_tuple = ("mouse", [8, 4, 6], (1, 2, 3))

# tuple can be created without parentheses
# also called tuple packing
my_tuple = 3, 4.6, "dog"
# tuple unpacking is also possible
a, b, c = my_tuple


my_tuple = ("hello")
print(type(my_tuple))  # <class 'str'>

my_tuple = ("hello",)
print(type(my_tuple))  # <class 'tuple'>

my_tuple = "hello",
print(type(my_tuple))  # <class 'tuple'>
```

```
<class 'str'>
<class 'tuple'>
<class 'tuple'>
```

Accessing Elements in a Tuple

Indexing

```
my_tuple = ['p', 'e', 'r', 'm', 'i', 't']

print(my_tuple[0])
print(my_tuple[5])

# Index must be in range
# These lines will produce errors if uncommented
# print(my_tuple[6])    # IndexError: list index out of range
# print(my_tuple[2.0])  # TypeError: list indices must be integers, not float

n_tuple = ("mouse", [8, 4, 6], (1, 2, 3))

print(n_tuple[0][3])  # Nested index
print(n_tuple[1][1])  # Nested index
print(n_tuple[2][0])  # Nested index
```

```
p
t
s
4
1
```

Negative Indexing

```
my_tuple = ['p', 'e', 'r', 'm', 'i', 't']

print(my_tuple[-1])
print(my_tuple[-6])
```

```
t
p
```

Slicing

```
my_tuple = ('p', 'r', 'o', 'g', 'r', 'a', 'm', 'i', 'z')

print(my_tuple[1:4])   # elements 2nd to 4th
print(my_tuple[:-7])   # elements beginning to 2nd
```

```
print(my_tuple[7:])    # elements 8th to end
print(my_tuple[:])     # elements beginning to end
```

```
('r', 'o', 'g')
('p', 'r')
('i', 'z')
('p', 'r', 'o', 'g', 'r', 'a', 'm', 'i', 'z')
```

Changing or Deleting a Tuple

```
my_tuple = (4, 2, 3, [6, 5])

# We cannot change an element of a tuple
# my_tuple[1] = 9
# TypeError: 'tuple' object does not support item assignment

# my_tuple[3] = 9
# TypeError: 'tuple' object does not support item assignment

# But item of mutable element can be changed
my_tuple[3][0] = 9

print(my_tuple)

# Tuples can be reassigned
my_tuple = ('p', 'r', 'o', 'g', 'r', 'a', 'm', 'i', 'z')

print(my_tuple)
```

```
(4, 2, 3, [9, 5])
('p', 'r', 'o', 'g', 'r', 'a', 'm', 'i', 'z')
```

Tuple Concatenation and Repetition

```
# Tuple Concatenation
print((1, 2, 3) + (4, 5, 6))

# Tuple Repetition
print(("Repeat",) * 3)
```

```
(1, 2, 3, 4, 5, 6)
('Repeat', 'Repeat', 'Repeat')
```

keyword del

```
my_tuple = ('p', 'r', 'o', 'g', 'r', 'a', 'm', 'i', 'z')

# Can't delete individual items from a tuple
# del my_tuple[3]
# TypeError: 'tuple' object doesn't support item deletion

# Can delete entire tuple
del my_tuple

# Accessing deleted tuple will produce NameError
# print(my_tuple)
# NameError: name 'my_tuple' is not defined
```

Python Tuple Methods

```
my_tuple = ('a', 'p', 'p', 'l', 'e',)

print(my_tuple.count('p'))
print(my_tuple.index('l'))
```

```
2
3
```

Tuple Membership Test

```
my_tuple = ('a', 'p', 'p', 'l', 'e',)

print('a' in my_tuple)
print('b' in my_tuple)
print('g' not in my_tuple)
```

```
True
False
True
```

Iterating Through a Tuple

```
for name in ('John', 'Kate'):
    print("Hello", name)
```

```
Hello John
Hello Kate
```